

Java Code

↓  
compile

Bytecode

-JAR

→ client

-JAR v/d

↓  
Bytecode

↓  
JVM (JIT)

↓  
machine  
code  
→ CPU

• net code

↓

compile code  
↳ language compiler

↓

MSIL (Microsoft  
Intermediate  
language)

Assembly

• exe

• dll

↳ client

→ .NET (with MSIL)

↓  
CLR (Common Language  
Runtime)  
(JIT)

↓  
Binary / machine  
code → CPU



VB-net

dim i as integer (VB)

↓  
vbc.exe (VB compiler)

↓  

System.int32 i

 MSIL .exe  
dll

↓  
CLR (JIT) → machine code → CPU

| C++          | Range                | MSIL (Common Type)  |
|--------------|----------------------|---------------------|
| <u>int</u>   | min - max<br>$x - y$ | <u>System.int32</u> |
| <u>short</u> | min - max<br>$p - q$ | <u>System.int16</u> |
| <u>long</u>  | min - max<br>$r - s$ | <u>System.int64</u> |

(CTS) Common Type System

IACS0 # C1 features specified by  
(common language specification)

digit i #

• IACS0

CL6

read

compiler (IACS0 C-EXE)

refer (CTS)

write

system int 32 i

```
class SqlConnection
{
}
class SqlCommand
{
}
class SqlDataReader
{
}
```

```
class OleDbConnection
{
}
class OleDbCommand
{
}
class OleDbDataReader
{
}
```

# System.Data

## Namespace

SqlClient

SqlConnection

SqlCommand

SqlDataReader

Sql

Sql

namespace  
OLEDB

OLEDBConnection

OLEDBCommand

OLEDBDataReader

OLEDB

OLEDB

namespace ODBC

ODBCConnection

ODBCCommand

ODBCDataReader

ODBC

ODBC

System-Data-11

ns System-Data

ns SmlClient

Classes

ns OLEDB

Classes

ns ODBC

Class

Visual Studio Editor + Compiler + debugger

.net Framework

mscorlib.dll

System.dll

System.Xml.dll

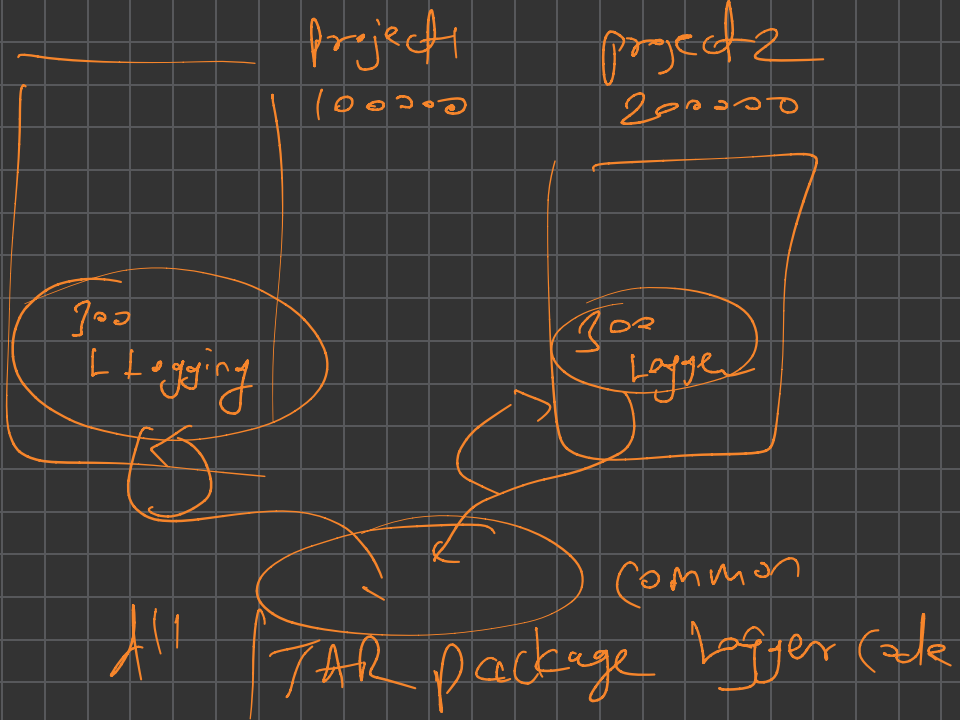
System.data.dll

System.——.dll

pos  
of dlls

System.——.dll





Solution

✓  
✓

Project 0 (Logger)  
├ Code  
└

Refer  
here →

Project 1  
├ Code  
└ Code

Project 2

